

INTERNET From The Sky

Unbounded by traditional ground infrastructure, satellite internet delivers broadband internet to places where access has been unreliable or completely unavailable. All that is needed is a small satellite dish to receive the signal and pass the bandwidth on to your router.

A new space race is on and this time it is all about the internet



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Space based internet systems have been in use since last many years, facilitated by geostationary satellites at a staggering 35,786km height above the equator that zip at about 11,000km per hour. Signals beamed from the satellites cover about one-third of the Earth's surface; hence three to four satellites can cover the entire planet.

However, a significant time lag, called latency, of about 600 milliseconds occurs between seeking the data and receiving it. Satellite internet using lower Earth orbit (LEO) satellites reduces the latency to 20-30 milliseconds, which is close to that of terrestrial systems.

To go around the Earth every few hours and balance the effect of gravity, the speed of LEO satellites should be more than double the speed of geostationary satellites, as a result their visibility at any location is reduced and

many satellites are needed to cover the complete surface of Earth. Starlink is expected to deploy as many as 42,000 satellites at altitudes of 350km to 1,200km.

The satellite internet provides consistent signals with high speed and low latency to preclude annoyances, such as

dropped calls and lost or delayed signals. Satellite internet is most suitable for areas where it is not feasible or viable to lay cables or set up transmission towers due to inaccessible terrain and high costs.

How satellite internet works

Satellite internet connections are provided by an array of interlinked small sized satellites moving in LEO. This array beams the internet signals down to transreceivers located on ground.

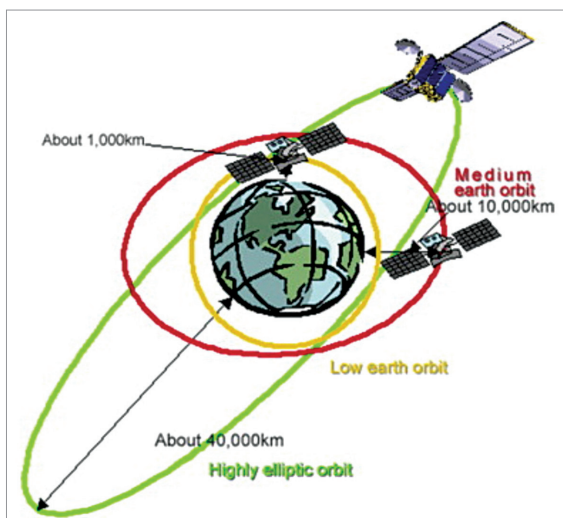
The signals are received by a dish antenna having an unobstructed access to the sky, installed in the premises of the user. When powered up using an electric connection, the self-pointing dish antenna scans the sky and locks onto the nearest overhead satellite, seamlessly maintaining the connection as each new satellite comes into view and the previous one fades beyond the horizon.

The user is connected to a ground station, called gateway (see box), located within about 800km to exchange data with the satellites. The user device is connected to the dish antenna through a modem-cum-router, called a user access terminal (UT).

The user's internet data request passes from the UT to the dish antenna and the dish antenna transmits the request to the satellite that beams it to gateway. The requested internet data is transmitted from the gateway to the satellite, and thereafter from satellite to the dish antenna, down to the UT, and finally onto the user device.

Advantages of satellite internet

Remote area access and high range are two major benefits, since satellite internet



LEO, MEO, and HEO orbit positions around Earth
(Source: <https://www.sciencedirect.com>)